

Sibankar Mandal

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Data Scientist with 2.7+ years of experience in **credit risk modelling** at ICICI Bank and currently at **Genpact** (Sep 2024 - present) specializing in **credit risk model (IFRS9, BASEL III, CECL) implementation and stress testing**. Proficient in **Python, R, SQL, and Big Data** technologies such as Hadoop, HDFS, PySpark, Apache Zeppelin, and YARN. Expertise in building **predictive models, scorecards**, optimizing financial processes, and managing large datasets to enhance customer scoring accuracy and risk management.

PROFESSIONAL EXPERIENCE

GENPACT

Data Scientist (Credit Risk model Implementation)

Kolkata, IN
Sept 2024-present

- Conducted **User Acceptance Testing (UAT)** for complex **credit risk models** under **IFRS 9** and **CECL** frameworks, ensuring testing plans aligned with **business requirements** and **forecasted credit losses**.
- Collaborated with **cross-functional teams** to perform **end-to-end testing, regression testing, production support, and implementation of performance enhancements** for **critical risk systems** generating **Risk-Weighted Assets (RWA)** and **loss forecasts**.
- Supported **Firm-wide Stress Testing (FWST)** processes under **IFRS 9** and **CECL** frameworks by validating model outputs of **stress testing models** to ensure accuracy and regulatory compliance.

ICICI BANK

Data Scientist (Credit Scoring & Risk Analytics)

Hyderabad, IN
June 2022 – July 2024

- Developed an **Exclusion-base** system targeting the generation of **Pre-Approved(PA)** offers for exclusive KTB (Known to Bank) customers are initially excluded from the PA process due to non-compliance with specific norms defined by ICICI Bank policies. Utilized the relaxation of predefined norms to generate offers, resulting in an additional uptake of approximately 15L for CD, 23L for TW, and 36L for PayLater products.
- Performed PL PA **Credit Test Experiment** using **Decision Tree** based algorithm, expecting of 3.5% marginal increase in monthly offer base count, planning to extend this experimentation to other products like CC, AL, HL.
- Build a **Logistic regression-based Scoring Model** using PayU scrubbed data to score NTB customers providing a marginal increase of 5L in PA offer base by augmenting some missing information like Application Score and Transaction Score.
- Developed a **Bureau Data-based Income Estimation Model** to predict the income of retail customers at the time of sourcing using bureau variables. Utilized the **XGBoost Machine Learning algorithm** in Python, achieving an accuracy of approximately 62.5% with a 15.33% overestimation rate.
- Developed a **Sourcing DataMart** for Credit Cards and Personal Loans, enabling comprehensive sourcing-level analysis for the data science and analytics department.

SKILLS

- Programming Languages:** **Python** (TensorFlow, Keras, PyTorch, Pandas, Scikit-learn, NumPy, Matplotlib, Seaborn, Pyspark), C, C++, R, **SQL**, BigQuery, HTML, CSS, SAS.
- Platforms:** Tableau, Google Cloud Platform (GCP), Azure, Microsoft Office, Excel, JIRA, Visual Studio Code, Vertica.
- Data Science & Miscellaneous Technologies:** ETL, Data science pipeline, Machine Learning, Deep Learning, Big data, Hadoop, Hive, Spark, Statistics, Time series, Experimental design, Hypothesis testing, OOP, APIs, Excel, Git.
- Certifications:** Microsoft Azure AZ-900, Machine Learning and Deep Learning in Python and R|Udemy, Tableau for Finance: Visualize Financial Data using Tableau |Udemy.

EDUCATION

NIT Trichy, Department of Computer Applications

Trichy, Tamil Nadu
June 2022

Masters in Computer Science

- Cumulative GPA: 7.84/10.0
- Relevant Coursework: Machine Learning, Data Science Statistics, Big Data Analytics, Probability & Discrete Mathematics

University of Calcutta, City College

Kolkata, WB
June 2019

Bachelor of Science in Mathematics

- Percentage: 6.88/10.0

PROJECTS

Fire Weather Index Prediction Web Application –(NIT TRICHY)

Aug 2021 - Dec 2021

- Developed and validated a robust Fire Weather Index (FWI) prediction model utilizing various **regression** techniques, achieving 93.02% accuracy with Ridge regression and cross-validation, based on weather data from Bejaia and Sidi Bel-abbes regions of Algeria during June to September 2012.
- Developed an interactive web application using **Python Flask** framework for FWI forecasting.

Stress Prediction of Working Employee Using ML Techniques –(NIT TRICHY)

Jan2022 - June 2022

- Developed a stress prediction model for working employees amidst the COVID-19 pandemic, aiming to **identify early signs of stress** for preemptive remediation, utilizing **Logistic Regression, Decision Tree, SVM, k-NN, and Random Forest**.
- Optimized models with **Grid Search**, achieving accuracies of 87.65%, 87.31%, and 86.80% for Logistic Regression, SVM, and Random Forest, respectively. Employed Recursive Feature Selection for enhanced stress detection.

ACHIEVEMENTS & RESPONSIBILITIES

- Qualified for Mathematics M.Sc. at the Tata Institute of Fundamental Research (TIFR) in 2019.
- Achieved an impressive All India Rank (AIR) 290 in the Joint Admission Test for M.Sc. (JAM-MS) in 2020.
- Demonstrated leadership as a Class Representative (CR) during Bachelor's degree program from 2016 to 2019.
- Contributed to campus welfare as a Hostel Member of the Mess Food Decision Committee in the academic year 2021-2022.